

Review for the First Midterm Exam

1 First Midterm Exam

- on Friday 2 October, in class or take home
- skipping policy

2 Overview and Review Questions

- convex hulls in the plane
- line segment intersection
- polygon triangulation
- linear programming
- orthogonal range queries

3 Fall 2023 First Midterm Exam

- five questions

MCS 481 Lecture 16
Computational Geometry
Jan Verschelde, 30 September 2026

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the first midterm exam on Friday 2 October

The exam covers the first five chapters:

- 1 convex hulls in the plane,
- 2 line segment intersection,
- 3 polygon triangulation,
- 4 linear programming,
- 5 orthogonal range queries.

There will be five questions, one of each chapter.

The review aims to prepare for the midterm exam, not to offer a complete coverage of all topics.

The exam is open book, open notes, and open computer, your answers must be uploaded into gradescope, by 12:50pm.

The take-home version of the exam will involve computer experiments; is due by noon on Monday 5 October 2026.

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skipping policy

About the policies in the course:

- Students may opt out of the exams and work on a computer project of their own research interest.
- In particular, concerning the exams:
If an exam is missed, then greater weight will be placed on the final exam,
especially on the material covered on the missing exam.
Students may replace the final exam with a research computer project combined with a written technical report.

Even if you would opt for a final project,
then taking the exam will give you homework credit.

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specification of input and output

slide 12 of lecture 1

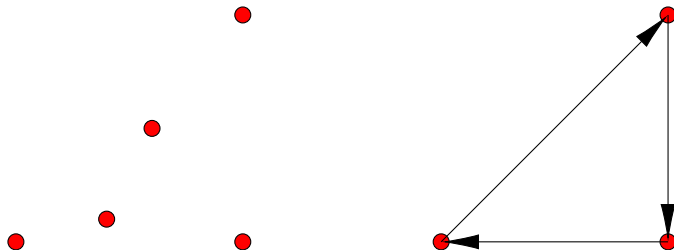
Input: n points in $\mathbb{R}^2$, $S = \{p_0, p_1, \dots, p_{n-1}\}$, $\#S = n < \infty$.

Output: list of points sorted in clockwise fashion

so two consecutive points in the list span an edge:

$L = [q_0, q_1, \dots, q_{m-1}]$, $(q_i, q_{i+1 \bmod m})$ spans an edge,
for $i = 0, 1, \dots, m-1$.

$S = \{(4, 1), (0, 0), (10, 0), (6, 5), (10, 10)\}$, $L = [(0, 0), (10, 10), (10, 0)]$



convex hulls in the plane

Given a finite set of points in the plane,
compute the convex hull as a list,
ordered so two consecutive points in the list span an edge.

Consider the following question.

Assume the n points in the plane is given on input
as a sorted list, *sorted on the y -coordinate*.

Explain how you can compute the convex hull in $O(n)$.

- 1 Describe the main idea in your algorithm.
- 2 Formulate pseudo code in sufficient detail
so you can show its correctness.
- 3 Show the running time of the algorithm is $O(n)$.

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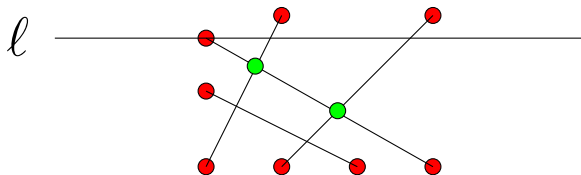
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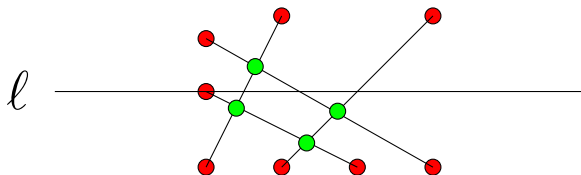
handling events

slide 16 of lecture 4

Invariant: all intersections with segments above ℓ are computed.



At the next event:



line segment intersection

Given a number of line segments, compute all intersection points.

Consider the following question.

What if all end points would lie on a circle with radius one?

In particular, the input is a sequence of angles (α_i, β_i)
and the line segments start at $(\cos(\alpha_i), \sin(\alpha_i))$
and end at $(\cos(\beta_i), \sin(\beta_i))$, for $i = 1, 2, \dots, n$.

- 1 Does this polar representation of the input simplify the problem?
- 2 Reformulate the plane sweep algorithm for *a rotating sweep line*.

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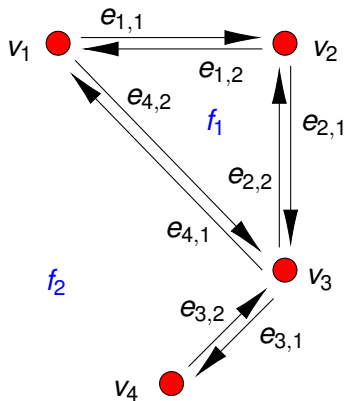
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recording vertices, half edges, faces – an example

slide 24 of lecture 5



polygon triangulation

Triangulating a convex polygon is straightforward.

We discussed the partitioning of a general, nonconvex polygon into y -monotone pieces.

Consider the following question.

Given a triangulation of a polygon stored as a doubly-edge connected list,

- 1 write pseudo code to compute *the area of the polygon*.
- 2 What is the cost of this algorithm?

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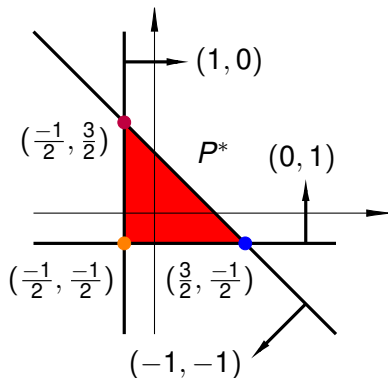
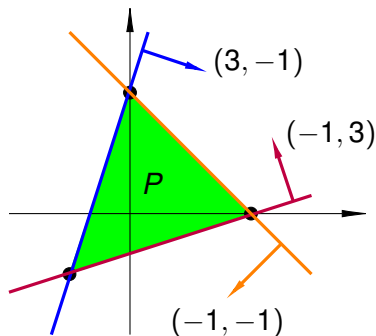
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duality or polarity (slide 4 of lecture 12)

P is defined by a finite set of half planes:

$$(-1)x_1 + (-1)x_2 \geq -2, \quad (-1)x_1 + 3x_2 \geq -2, \quad 3x_1 + (-1)x_2 \geq -2.$$

Let $(-1/2, -1/2)$, $(-1/2, 3/2)$, $(3/2, -1/2)$ as $(v_1/(-b), v_2/(-b))$, associated with (v_1, v_2) and b . The dual of P is P^* .



linear programming

A linear programming problem asks to optimize a linear function over a polyhedron, given as a list of linear inequalities.

The coefficients of the linear inequalities are components of normals, vectors that are perpendicular to the edges.

Consider the following question.

A *divide and conquer method* to solve a linear programming problem splits the number of constraints in two halves.

- 1 Describe the method in sufficient detail with pseudo code.
- 2 What is the cost of this method?

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orthogonal range queries

To efficiently report all points in a range, geometric data structures, Kd trees and range trees, generalize a balanced binary search tree.

Consider the following question.

Suppose we use a 2D range tree to represent pairs of data where one element of the pair belongs to a wide continuous range (e.g.: salary) and the other element is discrete (e.g.: marital status, number of dependents).

- 1 Does it matter for the complexity of the 2D range tree whether we take as x the discrete or the *continuous data type*?
- 2 Describe the influence of the range of values for x on the search time in a 2D range tree.

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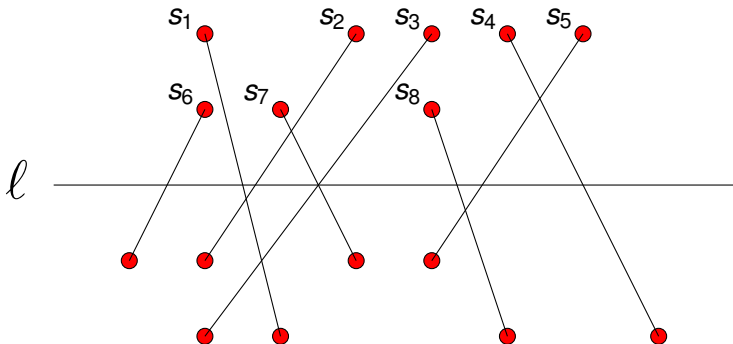
1. convexity

Consider $P = [(1, 0), (0, 3), (1, 2), (2, 3)]$ which represents a polygon with its vertices ordered in clockwise order.

Is P convex? Justify your answer.

2. intersecting line segments

Consider the segments $s_1 = ((1, 2), (2, -2))$, $s_2 = ((3, 2), (1, -1))$, $s_3 = ((4, 2), (1, -2))$, $s_4 = ((5, 2), (7, -2))$, $s_5 = ((6, 2), (4, -1))$, $s_6 = ((1, 1), (0, -1))$, $s_7 = ((2, 1), (3, -1))$, $s_8 = ((4, 1), (5, -2))$, intersected by the sweep line ℓ at height 0.



Draw the balanced binary search tree which represents the status.

3. doubly connected edge lists

Draw the doubly connected edge list for the nonconvex polygon spanned by the points listed in clockwise order:

$$[(3, 0), (0, 2), (3, 4), (2, 2)].$$

List the tables of vertex, edge, and face records.

4. duality

Construct the dual of the triangle spanned by the points $(2, 2)$, $(-2, 1)$, and $(0, -1)$ using inner normals to the edges.

5. Kd trees

Draw the kd tree for the points in the list

$[(3, 7), (0, 5), (8, 8), (2, 4), (6, 5), (9, 5), (8, 0), (3, 0)]$.

Observe that some points have the same x - and y -coordinate.

Your answer should reflect the method we have covered to deal with points which have the same coordinates.